A Hybrid Biomedical Image-Analysis Pipeline for Fluorescence-Microscopy Nuclei

Assignment 3 — AI in Biomedical Imaging | [Student Name] — [Student ID] | 17 August 2026

1 Overview of methods

Modality & data. The assigned modality is synthetic fluorescence microscopy (DAPI-like nuclear staining), 112 RGB 256×256 images with paired binary/instance masks, split 80/20/12 (train/val/test), plus 4 corrupted test variants for the robustness extension. All images were converted to 8-bit grayscale (PIL) and confirmed at 256×256 (bilinear resize kept in the code path for portability to other sizes).

Pipeline, as implemented. *Task 1* pairs an EDA (sample grid, pooled intensity histogram) with a direct image description from a local vision-language model (`llama3.2-vision` via Ollama), comparing a naive prompt against a schema-constrained, descriptive-not-diagnostic prompt, repeated 3× to expose run-to-run variability. *Task 2* segments the same image classically — Otsu threshold → morphological opening/closing & small-object removal → connected-component labelling → `regionprops_table` — turns the table into a short deterministic numeric summary, and sends *only that text* (never the pixels) to a smaller local text LLM (`llama3.2`) for a paragraph + JSON. *Task 3* trains a compact U-Net (4 encoder/decoder stages, base width 16, ≈1.94M parameters, skip connections, combined BCE+Dice loss) for 40 epochs (Adam, lr=1e-3, batch 4, CPU) and evaluates Dice/IoU on the held-out val split. *Task 4* chains the trained U-Net → regionprops → code-computed JSON fields → LLM narrative across all 12 unseen test images, aggregated to a CSV. Throughout, numeric JSON fields a user would act on are computed in code, never by LLM arithmetic — the design rationale behind this is discussed under Q4.

2 Task 1 — data preparation & VLM description

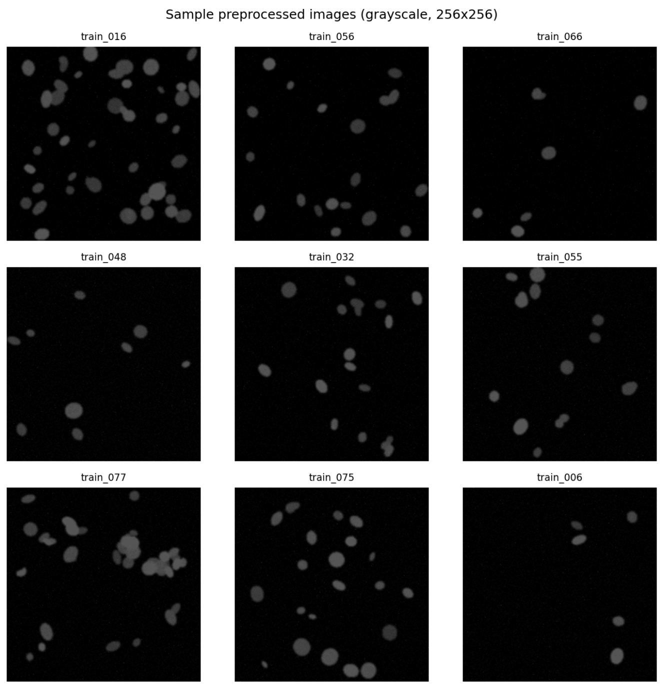

Fig. 1. Nine random preprocessed (grayscale, 256×256) training images.

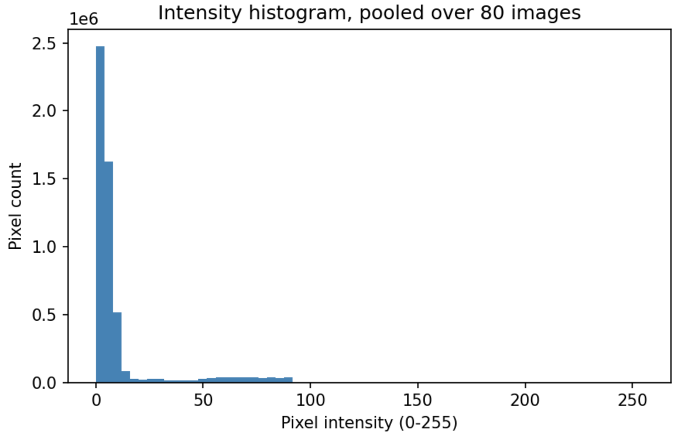

Fig. 2. Pixel-intensity histogram pooled over all 80 training images: strongly right-skewed, dark background dominates, with a long low-count tail from nucleus signal — typical of fluorescence images on a black field.

Naive prompt ("What is this image showing? ... is there anything wrong with the tissue?") returned free prose that volunteered an unprompted, confident clinical-sounding judgement: *"cells appear to be healthy and normal, with no visible signs of damage or abnormalities."* Nothing in the prompt asked for a health assessment — the model defaulted to one, which is exactly the failure mode Task 1 is designed to catch.

Optimised prompt (used for all reported VLM outputs)

```
You are an assistant that provides objective, descriptive
observations about a biomedical research image, for
educational purposes only. You are NOT a diagnostic tool:
you must not name a disease, give a diagnosis, or suggest
treatment. Describe only what is visually observable.

Report: modality, tissue_type, notable_features (1-2
sentences, purely descriptive), image_quality
(focus/contrast/noise). If not confident about any field,
write "uncertain" instead of guessing.

Respond with ONLY a single JSON object:
{"modality": "...", "tissue_type": "...",
"notable_features": "...", "image_quality": "..."}
```

(Full prompt text, incl. exact schema line, in `src/vlm_description.py` / `outputs/json_records/task1_vlm_description.json`; shown condensed here for space.)

```
{"modality": "fluorescence microscopy",
 "tissue_type": "uncertain",
 "notable_features": "Bright blue spots are scattered
   across the image, possibly representing cellular
   structures or other biological features.",
 "image_quality": "uncertain"}
```

Run-to-run variability. Three repeats of the identical optimised prompt on the same image gave different answers for `tissue_type` ("uncertain", "cell culture", "uncertain") and `image_quality` ("uncertain" twice, but once a full descriptive sentence). Wording of `notable_features` also changed each time. This is expected LLM sampling variance, not a bug — it is exactly why a single free-text VLM call should never be the sole record of an image (see Q1/Q4).

3 Task 2 — classical features & numbers-first interpretation

On `train_004`, Otsu + morphology detected 37 objects. A sample of the region table:

label	area	ecc.	solidity	mean int.
1	267	0.761	0.950	66.8
3	478	0.821	0.746	64.8
4	2036	0.826	0.565	73.5
7	102	0.650	0.953	52.0

Numbers-first interpretation prompt

```
You are an assistant that interprets a table of image-
analysis measurements... You are NOT given the image
itself, only these numeric measurements, and must not
invent visual details... You are NOT a diagnostic tool.

Measurements summary: {n_objects,
area/eccentricity/solidity/mean_intensity/extent stats}

1. Write ONE short paragraph (3-4 sentences) on count,
density, size spread, shape regularity.
2. Output JSON: {"n_objects": int, "density_class":
"sparse|normal|dense",
   "shape_regularity": "regular|irregular", "quality_flag":
"ok|uncertain"}
If ambiguous, use "uncertain" rather than guessing.
```

```
LLM json:  {"n_objects": 37, "density_class": "sparse",
            "shape_regularity": "irregular",
"quality_flag": "uncertain"}
Code json: {"n_objects": 37, "density_class": "normal",
            "shape_regularity": "irregular",
"quality_flag": "ok",
            "mean_area": 360.9}
```

Note the disagreement: the code rule (<15 sparse, 15–40 normal, >40 dense) puts 37 objects at *normal* density; the LLM, given that same count in the prompt text, wrote *sparse* — a numeric-reasoning slip even in a numbers-only, image-free setting. This is direct evidence that the code-derived JSON, not the LLM's JSON, must be treated as ground truth (Task 4 architecture; Q4).

4 Task 3 — U-Net segmentation

The U-Net (Section 1 architecture) was trained for 40 epochs on the 80-image train split and evaluated on the 20-image val split; Fig. 3 shows the resulting training and validation curves.

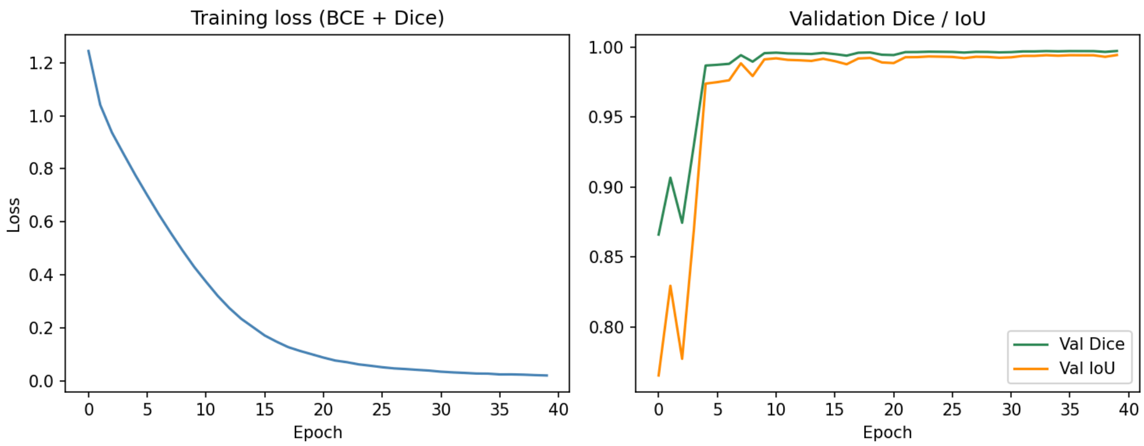

Fig. 3. Training loss (BCE+Dice) and validation Dice/IoU over 40 epochs. Both curves plateau by ~epoch 15-20; the visible dip in early-epoch Dice/IoU (epochs 1-3) reflects the randomly-initialised decoder before it learns to close small holes.

Split	Method	Mean Dice	Mean IoU
val (n=20)	Otsu (classical)	0.976	0.953
val (n=20)	U-Net (BCE+Dice, 40 ep.)	**0.997**	**0.994**

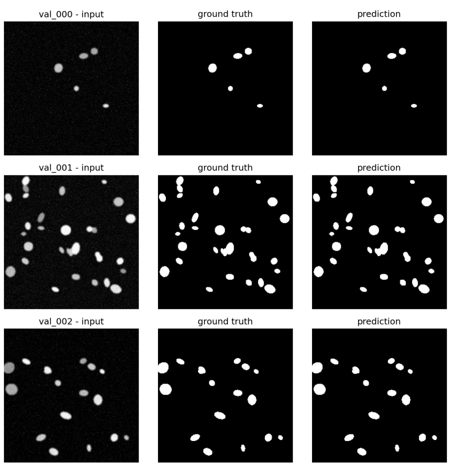

Fig. 4. Input / ground-truth / U-Net prediction for 3 validation images (val_000–val_002) — predicted masks are visually close to pixel-identical to ground truth.

5 U-Net vs Otsu, and the hybrid pipeline (Task 4)

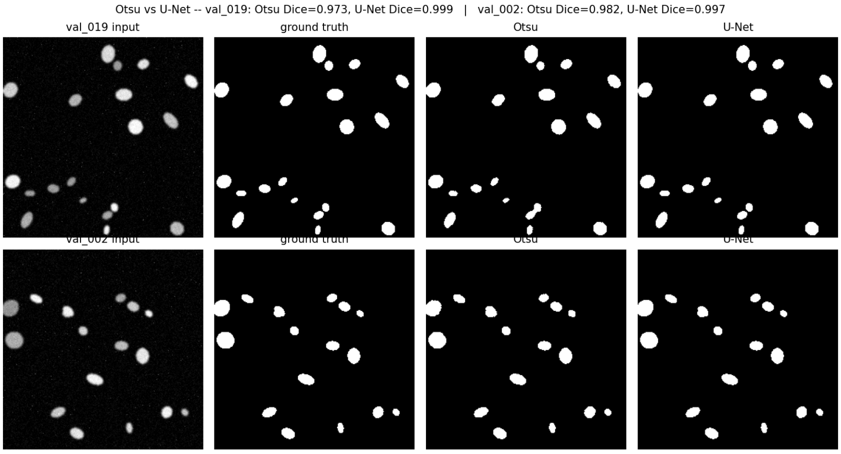

Fig. 5. Qualitative comparison on the two extreme cases from val: U-Net's largest margin (val_019, touching/medium-contrast nuclei — Otsu leaves ragged, notched edges) and its smallest margin (val_002, large well-separated bright nuclei — a global threshold is already nearly correct).

The U-Net matched or beat Otsu on *every* validation image; there was no case where Otsu was actually better, only cases where its disadvantage was small. That the gap is largest on medium-contrast/touching objects and smallest on large, bright, well-isolated ones (Fig. 5) is consistent with Otsu's single global threshold being a good approximation only when foreground/background separation is already easy.

Task 4 example record (test_000, unseen)

```
{"image_id": "test_000", "n_objects": 8, "mean_area":
191.0,
 "density_class": "sparse", "quality_flag": "ok",
 "narrative": "The test_000 image contains 8 objects with
   varying sizes, as indicated by a standard deviation of
   111 pixels in their areas. These objects appear to have
   a range of shapes ... suggesting some elongation or
   irregularity. However, the mean solidity and extent
   values suggest that most ... are roughly convex and
   well-defined ..."}
```

All 12 test images were processed this way (U-Net mask → regionprops → code JSON fields → LLM narrative) and aggregated into `outputs/task4_hybrid_records.csv` (one row per image; `n_objects` ranged 8–43, `density_class` sparse/normal/dense throughout). The narrative prompt keeps Task 2's numbers-only, non-diagnostic framing but asks for prose only (the JSON fields are already fixed by code, so the LLM is not asked to repeat them):

```
You are an assistant that writes a short, objective
narrative for a biomedical research image, for educational
purposes only. You are NOT a diagnostic tool. You are given
only numeric measurements below, not the image itself -- do
not invent visual details beyond what the numbers imply.
Image id: {image_id}
Measurements from automatic segmentation: {summary_text}
Write ONE short paragraph (3-4 sentences) ... If the
numbers are ambiguous, say so rather than guessing. Plain
text only, no JSON.
```

Robustness extension (bonus)

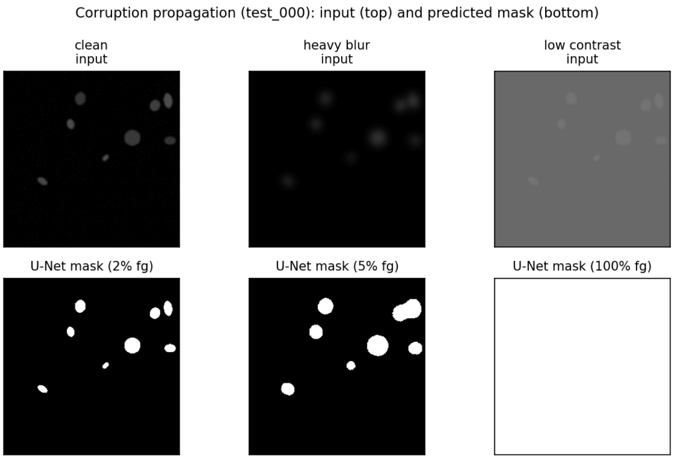

Fig. 6. Corruption propagation on `test_000`. Blur softens object boundaries but the mask still finds discrete objects. Low contrast is catastrophic: the U-Net mask becomes 100% foreground (the whole frame), silently breaking every downstream stage.

Two corrupted variants (heavy Gaussian blur; low-contrast rescale) of 2 test images were traced through all four stages. **Blur**: pixel std drops ~40–44% (stage 1), mask Dice-vs-GT collapses from ~0.997 to ~0.65 (stage 2, Fig. 6 middle column), object count drops modestly. **Low-contrast**: pixel mean shifts by hundreds of percent (stage 1), the mask *completely fails* to 100% foreground (Dice≈0.05–0.36, Fig. 6 right column), and the region table collapses to a single "object" with `area=65535` — exactly 256×256. Critically, the LLM narrative for this case still reads fluently ("a single, elongated oval-shaped object…") with no hint of failure — only the structured field `area=65535` exposes it. The earliest stage at which the corruption is *detectable at all* is stage 1 (raw pixel statistics, before segmentation even runs); the earliest stage at which it becomes *unambiguous without instrumenting raw pixels* is the stage-3 feature table, not the narrative.

6 Discussion & answers to the set questions

Q1 — VLM vs numbers-first: which is more useful, which is more trustworthy? The direct VLM description (Task 1) is more *useful* for a fast, human-readable first impression — it reads naturally and needs no downstream code to consume. It is less *trustworthy*: it is non-deterministic (Section 2), it is not grounded in any measurement that can be independently re-checked, and the naive-prompt run showed it will volunteer confident-sounding judgements it was never asked for. The numbers-first pipeline (Task 2) is more trustworthy because every quantity is reproducible — re-running Otsu+regionprops on the same image always gives the same table — and the LLM's role is reduced to phrasing rather than measuring. It is not perfectly trustworthy either (Section 3 showed a density-class misclassification), which is exactly why the *code-derived* JSON, not either LLM's JSON, is treated as ground truth throughout.

Q2 — Did the U-Net improve on Otsu? Examples. Yes, on every validation image (mean Dice 0.997 vs 0.976; mean IoU 0.994 vs 0.953). U-Net's advantage is largest on `val_019` (Otsu 0.973 vs U-Net 0.999) where nuclei are medium-contrast and partly touching — Otsu's fixed global threshold leaves ragged edges and small spurious fragments (Fig. 5, top row). Otsu is closest to U-Net (though still behind) on `val_002` (0.982 vs 0.997), where nuclei are large, bright, and well-isolated against a clean dark background — conditions under which a single threshold is nearly sufficient by construction.

Q3 — Dice/IoU meaning and failure modes. Dice = $2|P\cap G| / (|P|+|G|)$ and IoU = $|P\cap G| / |P\cup G|$ both measure pixel-set overlap between predicted (P) and ground-truth (G) foreground; IoU penalises boundary/size mismatches more harshly than Dice for the same error. Mean val Dice 0.997 / IoU 0.994 mean predicted masks are almost pixel-identical to ground truth on this dataset. The (narrow, 0.995–0.999) per-image range is lowest on images with more, smaller, or more closely packed nuclei (e.g. val_004, val_005, val_012) — consistent with the qualitative panels, where residual error concentrates at object boundaries and at touching instances. Because the model outputs one binary foreground mask rather than per-instance labels, it structurally cannot separate two touching nuclei into two objects even when the pixel-level Dice looks excellent — a real limitation Dice/IoU do not expose.

Q4 — Where can the LLM hallucinate, and what mitigates it? Three concrete points were observed: (i) Task 1's free-text `tissue_type`/`image_quality` fields changing answer across identical re-runs; (ii) Task 2's LLM assigning `density_class="sparse"` to 37 objects when the code rule (and its own stated count) implies "normal"; (iii) the robustness test's narrative describing a physically broken, whole-frame mask as "a single, elongated oval-shaped object" without flagging anything wrong. Mitigations used in the code: a constrained JSON schema that explicitly permits "uncertain" (removing the pressure to fabricate a confident answer); an explicit "not a diagnostic tool" role instruction; isolating the interpretation LLM from pixels entirely in Tasks 2/4 so it cannot invent visual detail beyond the numbers it is given; a regex-based JSON-extraction-with-retry helper (`llm_utils.chat_json`) that rejects and retries malformed output; and, most importantly, computing every numeric field a downstream user would act on (`n_objects`, `mean_area`, `density_class`, `quality_flag`) directly from the regionprops table in code, never from the LLM's own arithmetic. Keeping this code-derived JSON as the source of truth matters because it is deterministic and diffable (identical input → identical output, unlike the LLM text), can be automatically range-checked (e.g. flag `area` ≥ image area, exactly what would have caught the corrupted-image failure), and lets an auditor trace any number back to the exact pixels and function that produced it — the fluent narrative alone cannot offer any of that.

Q5 — Clinical trust and the highest-impact change. No part of this system should be trusted in a real clinical setting yet. The near-perfect Dice (0.997) reflects a fully synthetic dataset of clean, well-separated, uniformly-stained blobs on a flat dark background — the model has never seen real staining variability, overlapping nuclei at density, or acquisition artefacts, so the number says more about the dataset's ease than about the model's real-world skill. The VLM step is demonstrably non-deterministic and can produce confident-sounding but unverified claims. And evaluation is at the pixel/mask level only — there is no instance-level check, so the pipeline could report a plausible object count for the wrong reason. Considering accuracy, auditability, and dataset limits together, the single highest-impact change would be replacing the synthetic dataset with a real, expert-annotated microscopy/histology set (with a held-out clinical-site split) *and* wiring the automatic range/plausibility checks the robustness test motivates (e.g. reject or flag any code-derived record whose `area` is implausible, before any narrative reaches a user) — together these close both the generalisation gap and the "plausible but wrong" narrative failure mode this report demonstrates.

References

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